



Karnataka State Police
Government of Karnataka

Powered by **I2S**

Dataathon 2026

Technology partner



Team Details

- A. **TEAM NAME:** TEAM SLYTHERIN
- b. **Team leader name:** KABILAN K A
- c. **Team size:** 5
- d. **Problem Statement:** AI-Driven Crime Analytics & Visualization Platform



CORE IDEA

- Unifies **crime, case, evidence, and geographic data** in one intelligence platform.
- Converts fragmented records into **actionable insights** using **GIS, analytics, prediction, and MADHUKAR AI Copilot**.
- Supports faster movement from **statewide trends** to **district hotspots and individual cases**.

NOVELTY

- Combines GIS intelligence, crime analytics, investigations, and AI in one unified command dashboard.
- Enables seamless drill-down from State → District → Police Station → FIR/Case → Evidence.
- Uses Zoho Catalyst serverless services, QuickML risk prediction, and MADHUKAR AI Copilot for faster intelligence-led decisions.

INNOVATION & UNIQUENESS

Command Dashboard

Live KPIs, trends, alerts, and serious cases.

GIS Intelligence

Crime maps, hotspots, and district risk analysis.

Investigation Intelligence

Links cases, suspects, evidence, and timelines.

Secure AI-Driven Architecture

Catalyst APIs, QuickML, and MADHUKAR AI Copilot.



Why Our Solution ?

- Unlike task-specific systems, our platform **unifies crime monitoring, GIS analysis**, investigation, predictions, reporting in one interface.
- Predictive intelligence enables **proactive policing** instead of reactive decision-making.
- **Connected crime data** links cases, suspects, evidence

Automates Complex Crime Data Analysis



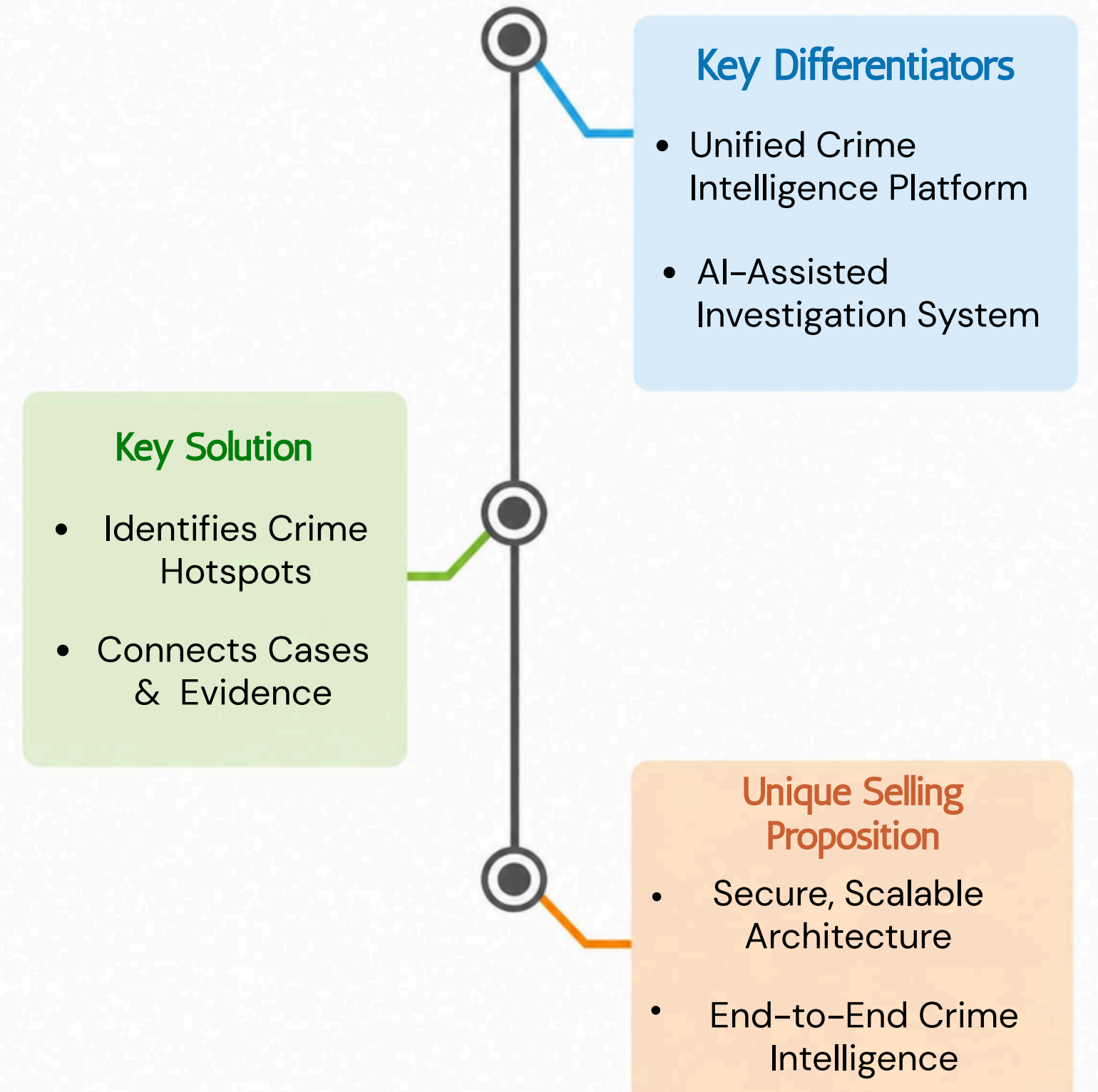
Enables Intelligence-Led Police Decisions



Accelerates Case Investigation and Resolution



Solution Impact

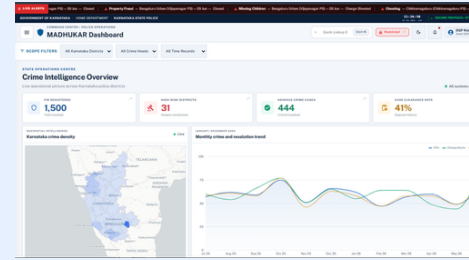




Key Features & Strategic Impact

Unified Command Dashboard

- Displays statewide KPIs, crime trends, district performance, alerts, and serious cases for rapid monitoring.



GIS & Predictive Intelligence

- Identifies crime hotspots, regional risks, emerging patterns, and anomalies through maps and predictive analytics.



Connected Investigation

- Links cases, suspects, evidence, timelines, and criminal networks to support faster investigations.



Social Benefits



- Strengthens Citizen Confidence
- Enables Proactive Policing
- Improves Response Planning



Operational Benefits



- Accelerates Investigations
- Optimizes Police Resources
- Reduces Manual Analysis
- Improves Operational Efficiency

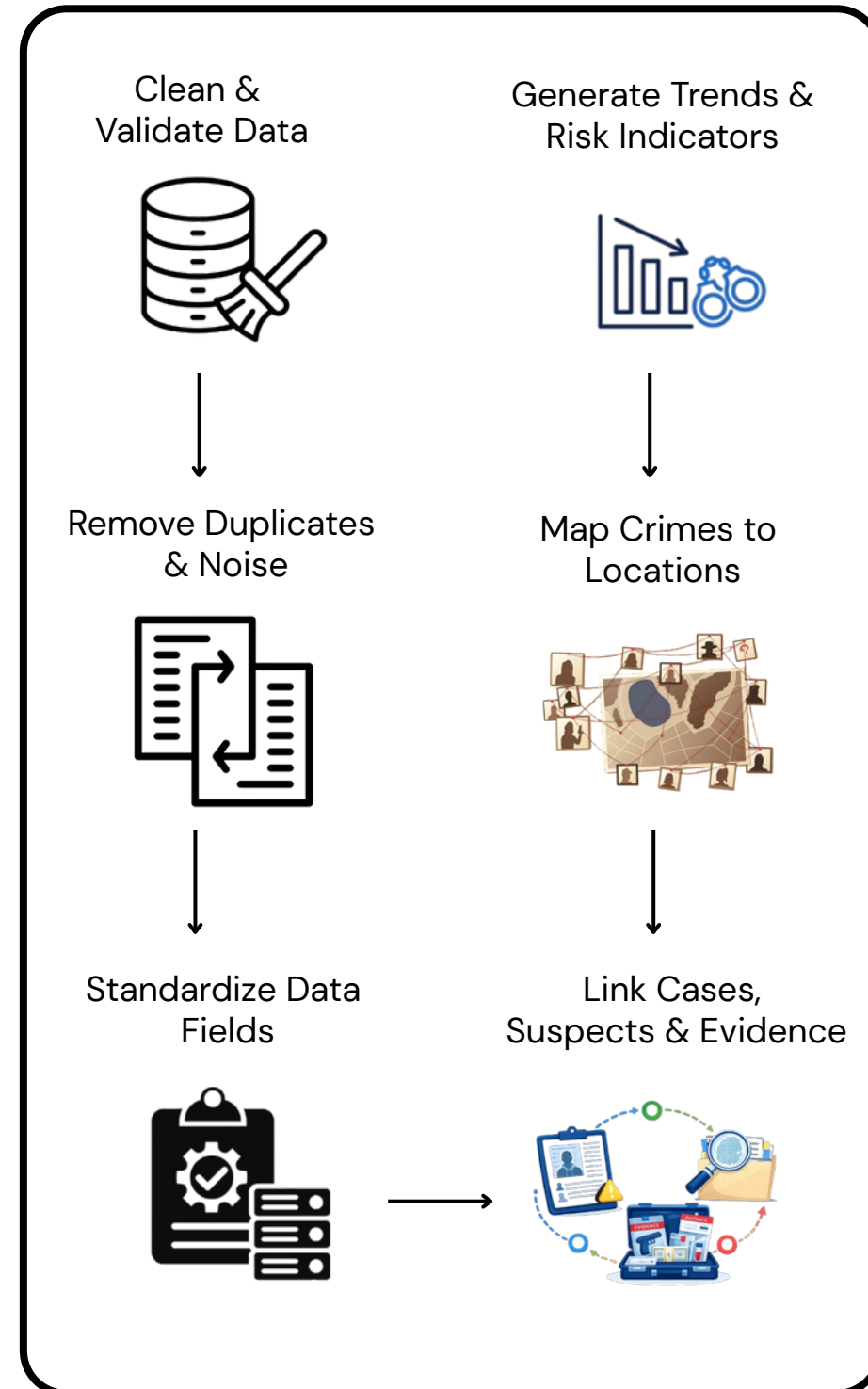


Process flow diagram

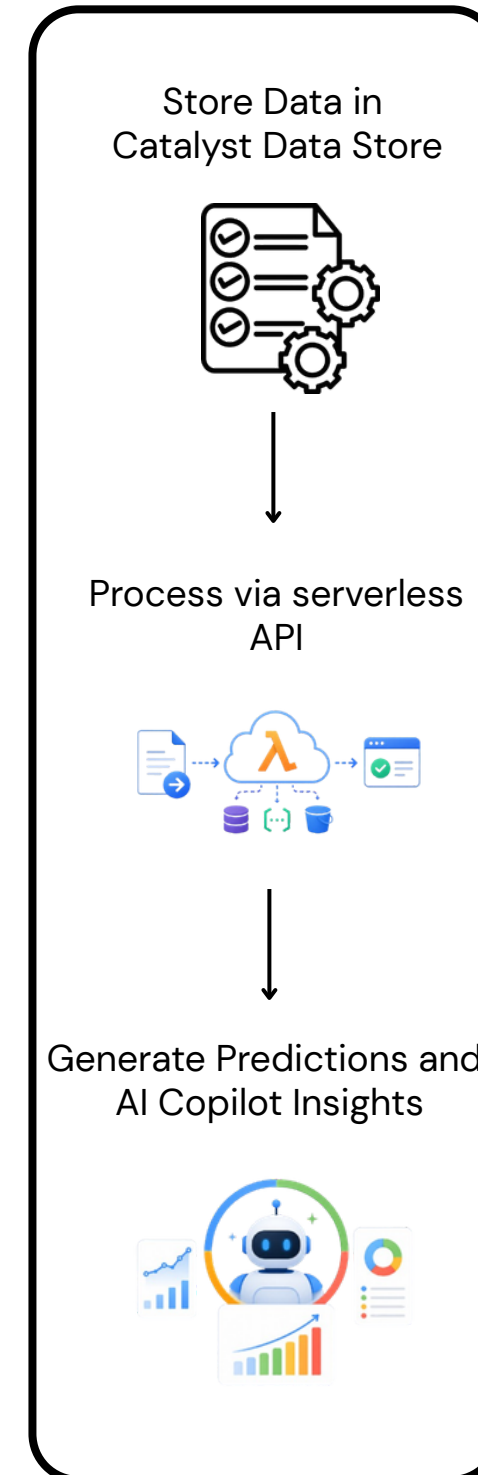
Data Input



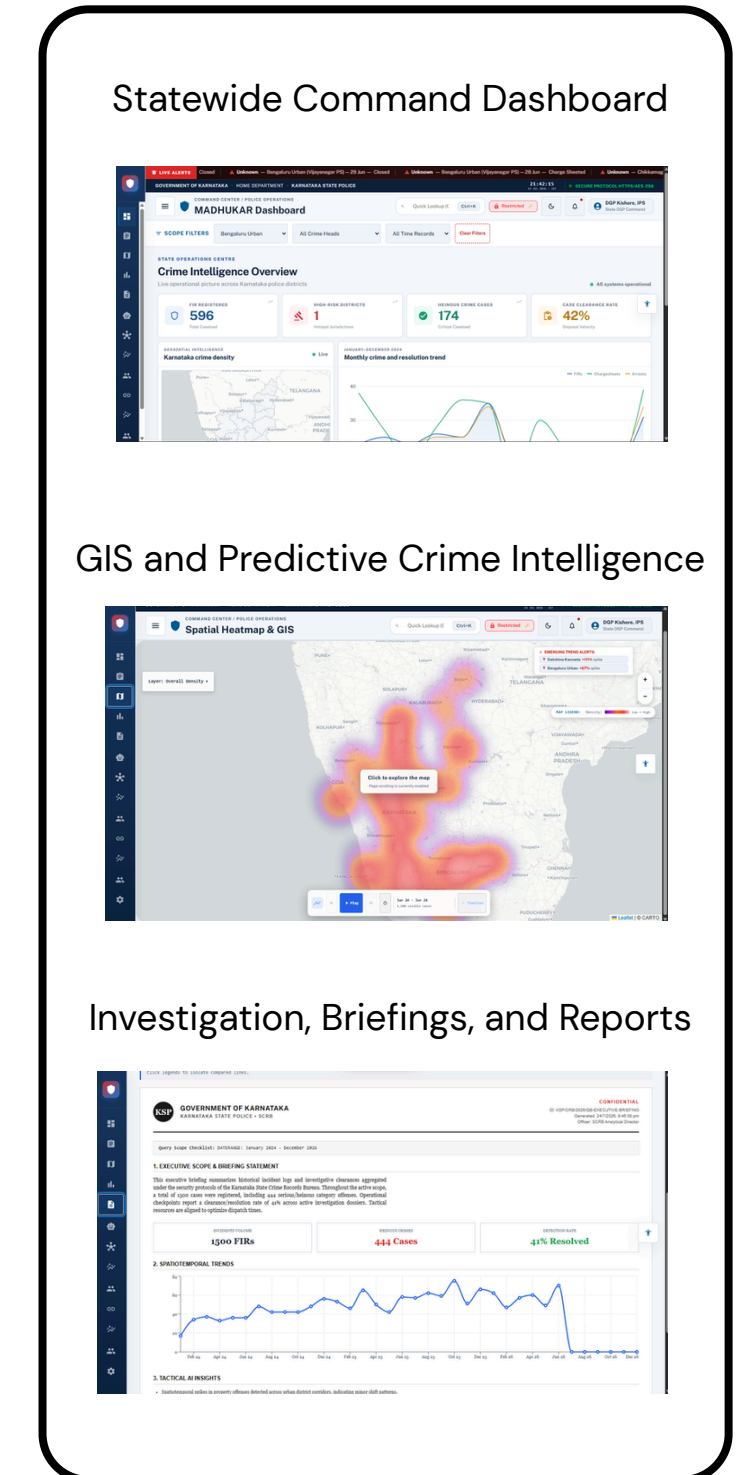
Data Preprocessing



Backend

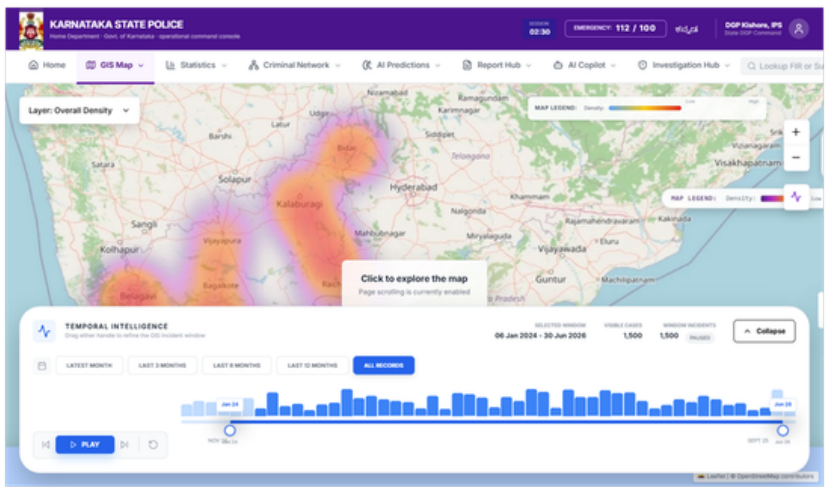


Output

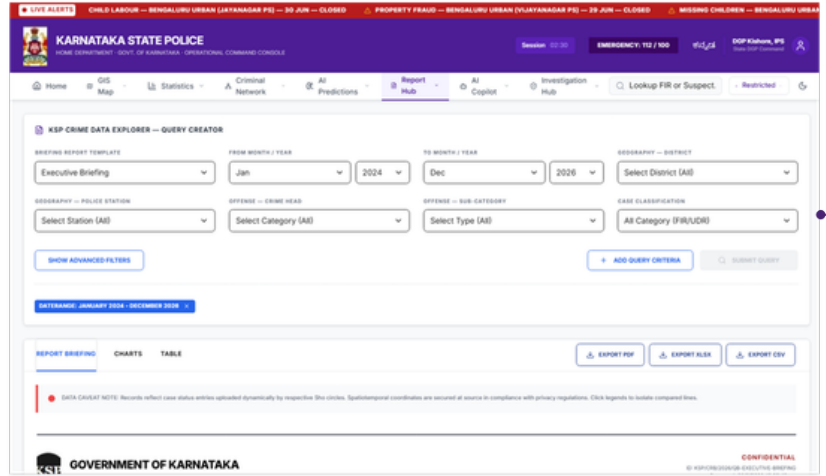


Mock Diagram

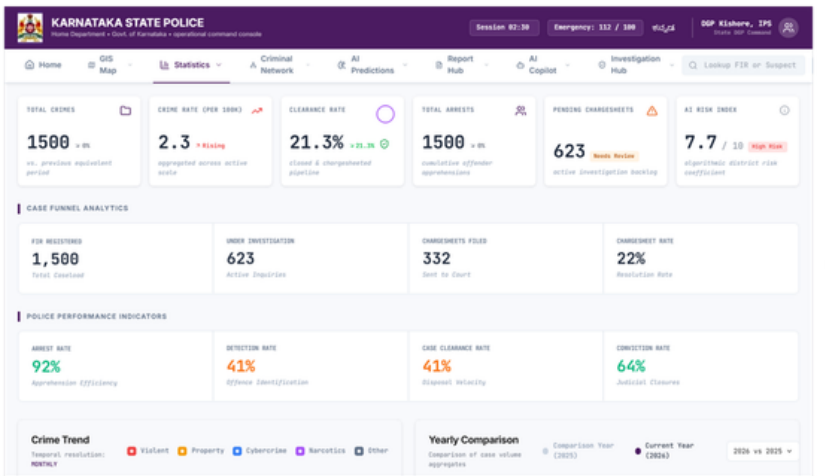
- Real-Time Command Center
- Geographic Crime Intelligence
- Crime Analytics & Trend Monitoring
- AI-Driven Intelligence
- Unified Investigation Workspace



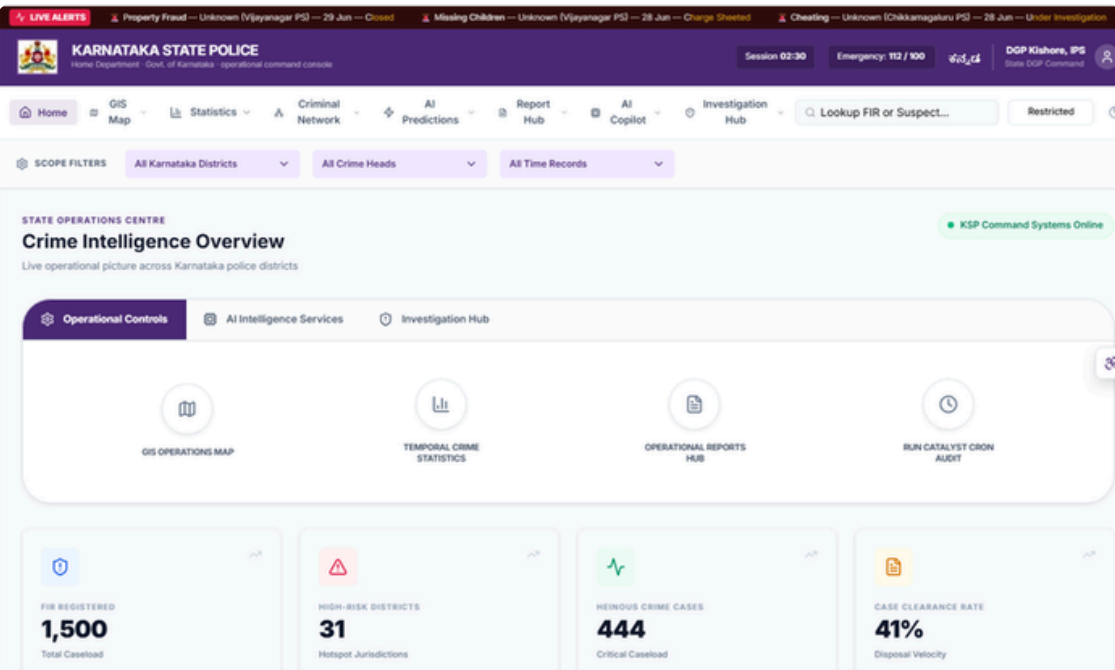
GIS MAP



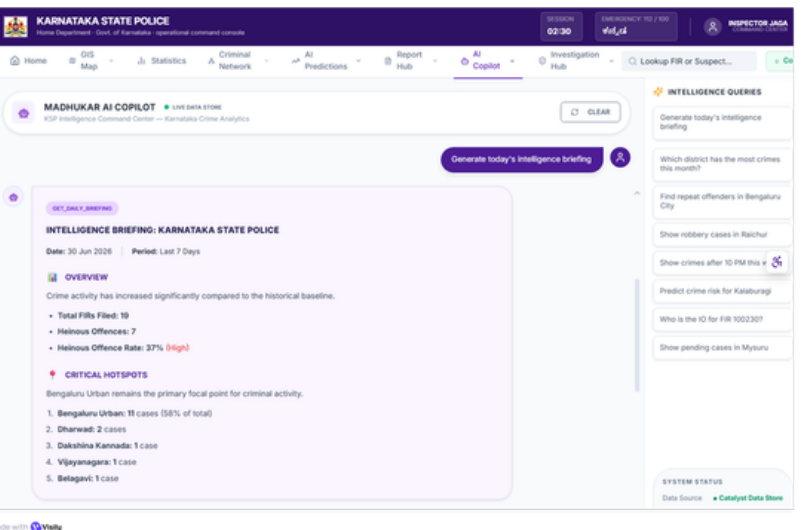
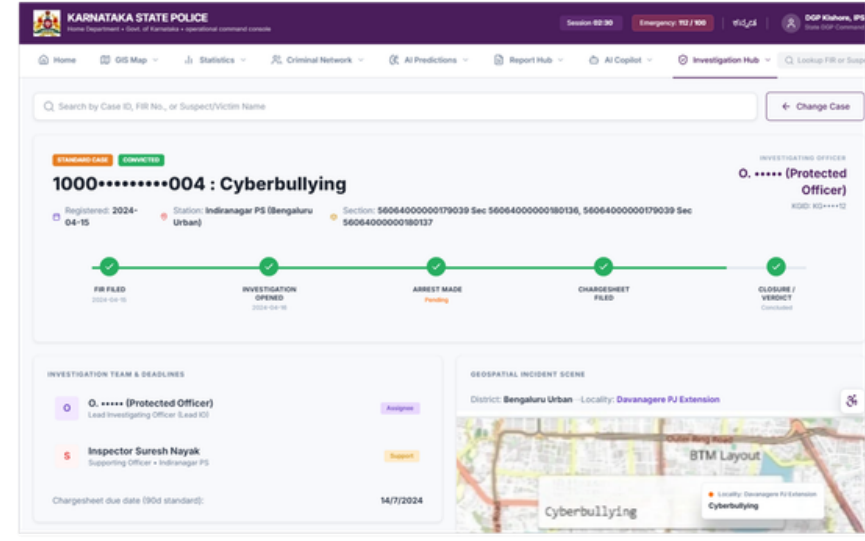
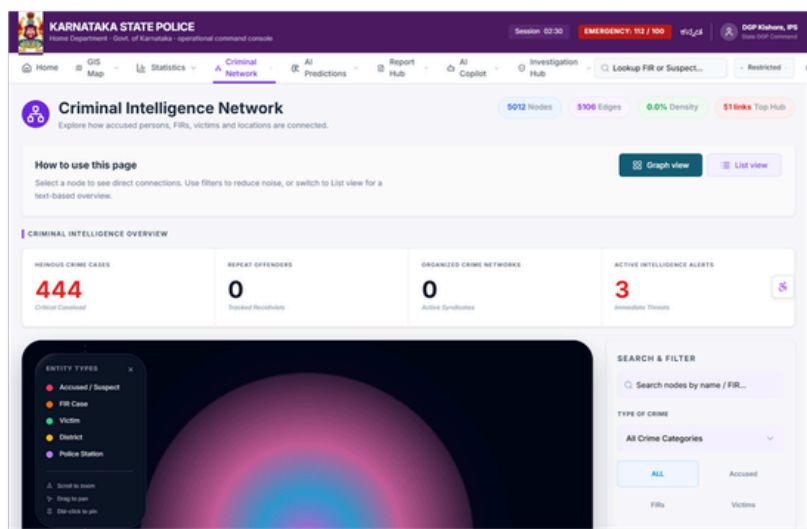
REPORT



STATISTICS

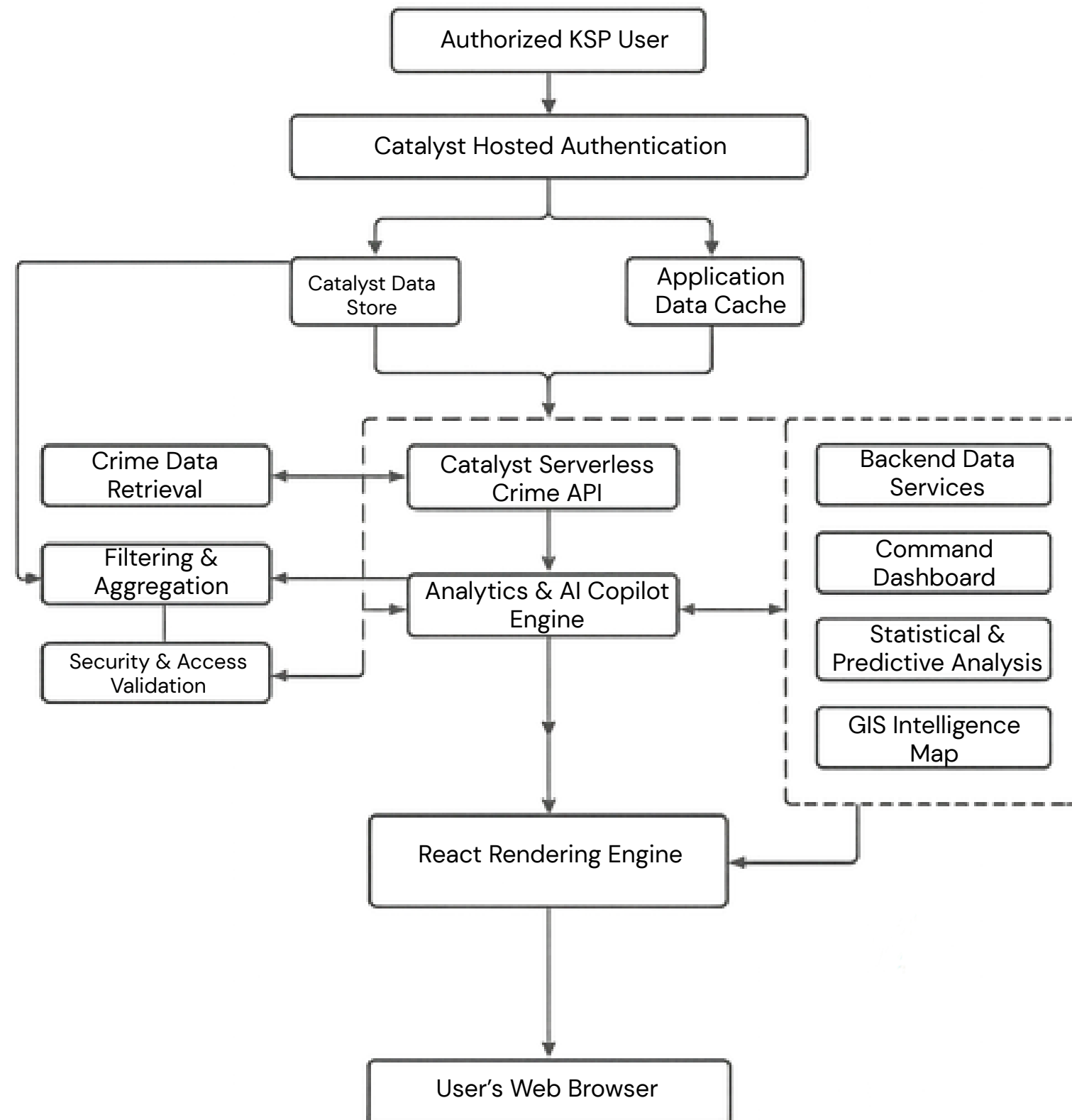


DASHBOARD

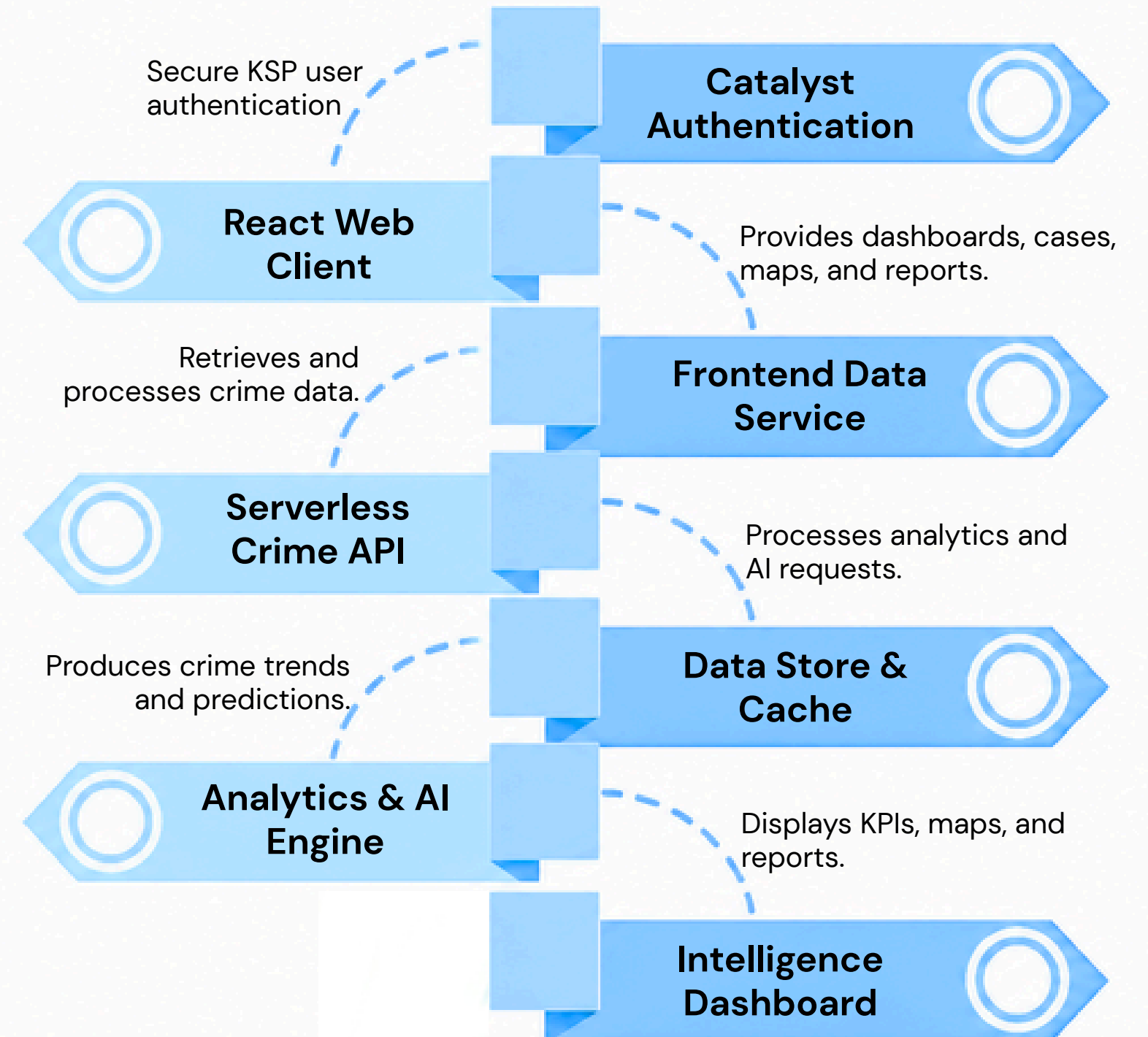




System Architecture



Component Workflow





Core Technology Stack

TECH STACK

Frontend

React 19



React Router



Recharts



Backend

Node js



Express js



Zoho Catalyst



Data & Geospatial Analytics

Catalyst Data Store



ZCQL



Leaflet



AI & Reporting

Catalyst QuickML + GLM-4.7-Flash



AI Copilot Engine



html2canvas + jsPDF



Operational Breakdown

Unified dashboard for KPIs, crime trends, alerts, and GIS maps.



Catalyst Advanced I/O API processes case data, analytics, predictions, and Copilot queries.



Catalyst Data Store + Cache manage police records, alerts, and OTP workflows.



QuickML and MADHUKAR AI generate risk predictions and intelligence insights.





Catalyst Services

CATALYST HUB

Catalyst Data Store

- Core relational database with 26 structured tables.
- Stores 1,501 active FIR crime records.

ZCQL Engine

- Executes dynamic SQL queries.
- Supports real-time date-range filtering .

Web Client Hosting

- Hosts the interactive Crime Analytics Dashboard.
- Visualizes crime network graphs.

Serverless Functions

- Powers the central Node.js Express crime API.
- Delivers pre-computed crime statistics

QuickML & GLM

- Serves as the core AI/ML framework
- Powers the MADHUKAR AI Copilot

OAuth & Cred Vault

- Provides primary authentication and security framework.
- Uses multi-strategy token authentication.

Cron / Schedulers

- Triggers automated background threat assessments and investigation audits.

Cache

- In-memory storage of daily threat alerts
- Secure, ephemeral storage of Multi-Factor Authentication

Catalyst Mail

- Used within the "Break-Glass" PII Verification flow
- email 6-digit OTP codes to authenticated officer's mail

Future Expansion Roadmap

- Role-Based Access Control (RBAC)
- Rank-Based Login (Commissioner, SP & Station IO)



- Secure Digital Evidence Vault
- Stores & Previews FIRs, Case PDFs, Weapon and Crime Scene Images



- Extracts Handwritten Text using Zia OCR
- Identifies Suspects & Locations with Zia NLP



- Automated UTC Risk Recalculation
- Real-Time SMS Alerts for Heinous Crimes



- Distributed Redis Caching
- Sub-Millisecond Query Performance

Estimated implementation cost

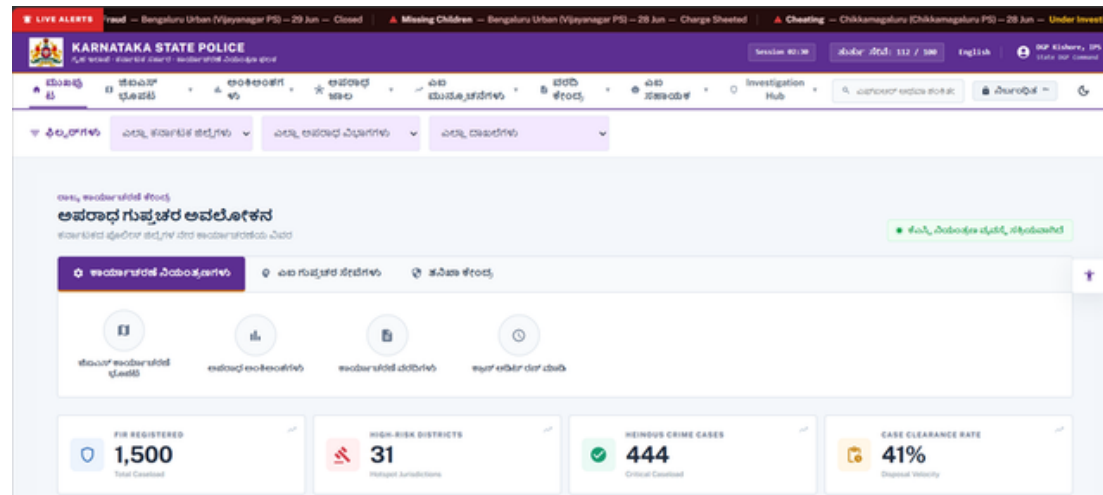
Dashboard Component	Estimated Cost
React Dashboard & UI/UX	₹1.5–2.5 lakh
Catalyst Backend & Serverless APIs	₹1.5–2.5 lakh
Data Store, ZCQL & Data Integration	₹1–2 lakh
GIS Maps & Spatial Analytics	₹1–1.8 lakh
Crime Prediction Models & AI Copilot	₹1.5–3 lakh
Security, Authentication & Access Control	₹0.8–1.5 lakh
Reports, Testing, Training & Deployment	₹0.8–1.5 lakh
Total Estimated Implementation	₹8.1–14.8 lakh

Catalyst Operating Cost

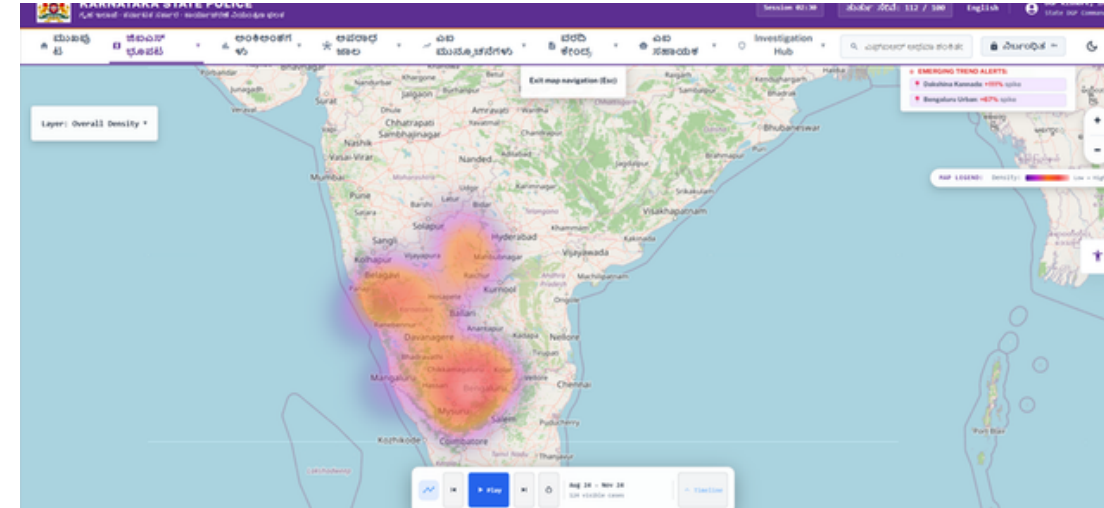
Deployment Stage	Recommended Plan	Estimated Cost
Hackathon Prototype	Development Environment	Free
Small Police Pilot	Pay-as-you-go	Based on actual usage
Controlled Production Pilot	Basic Subscription	USD 25/month
Multi-District Deployment	Standard Subscription	USD 50/month
Statewide High-Usage Deployment	Premium/Elite or Enterprise	USD 75–500+ per month
Maintenance & Technical Support	Development team	₹1.2–3 lakh/year

Disclaimer: Costs are indicative estimates for the current crime analytics dashboard scope. Actual implementation and operating costs will vary with number of users, crime-record volume, API calls, storage, map usage, AI model usage, security requirements, and third-party integrations

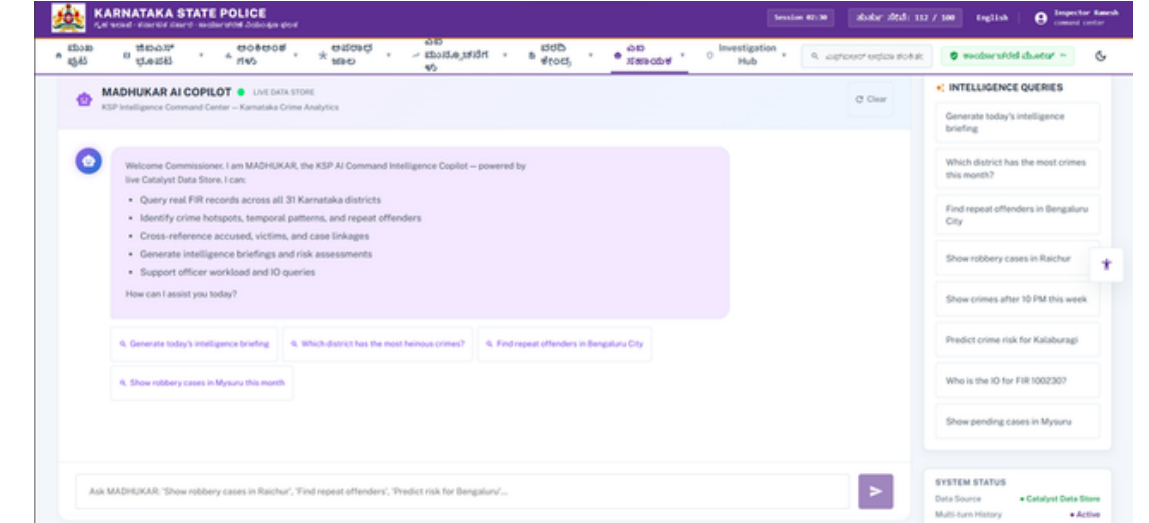
Snapshots of Prototype



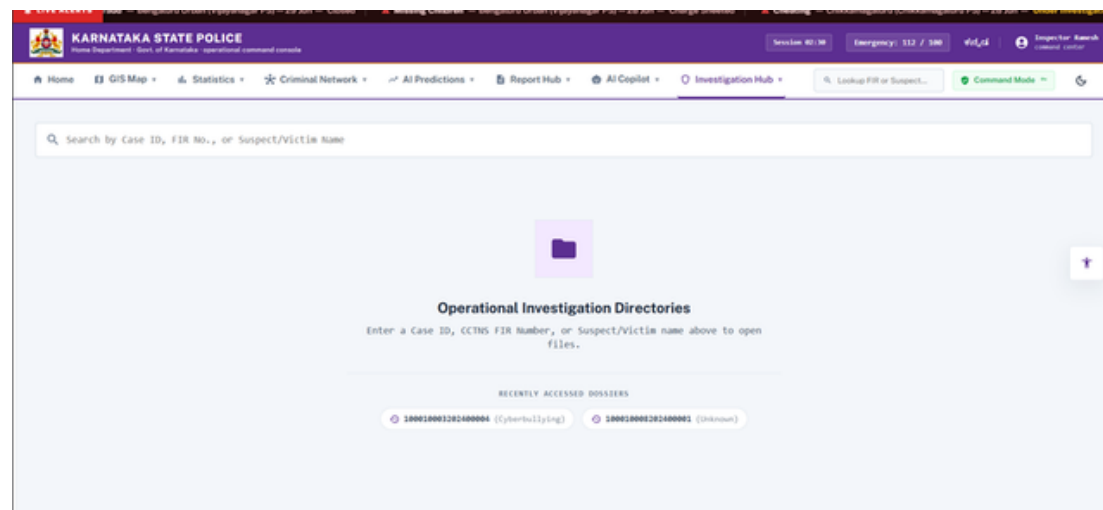
Dashboard



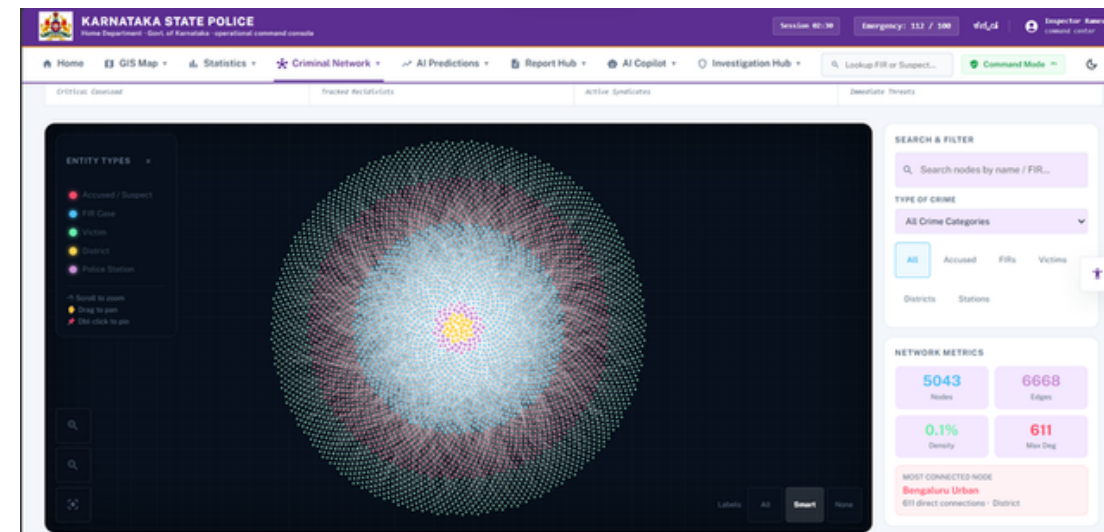
HeatMap



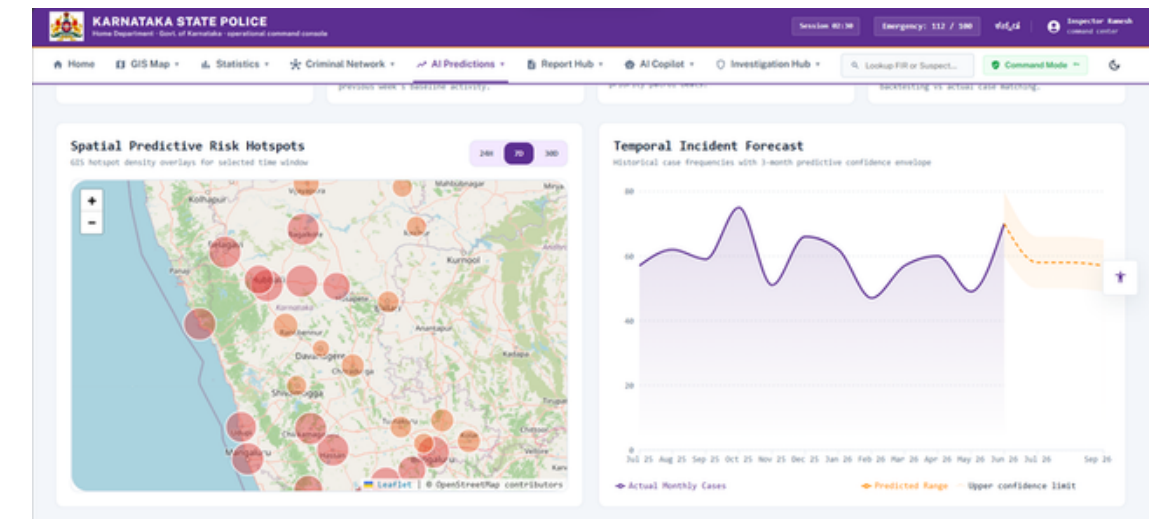
AI Copilot



Case Overview



Criminal Network



Predictions



Prototype Performance report/Benchmarking

Frontend Web Interface



928 ms

First Contentful paint

- 74/100 Lighthouse Score with First Contentful Paint optimized score

Database & GIS Mapping



366 ms

Response Time

- Instant Karnataka.geojson boundary response time.

AI Insight Engine



6.5 s

Response Time

- Stable multi-hop cloud generation for complex model queries.

Load Capacity & Stress Testing

~8.3

Request/Sec

0%

Error Rate

5

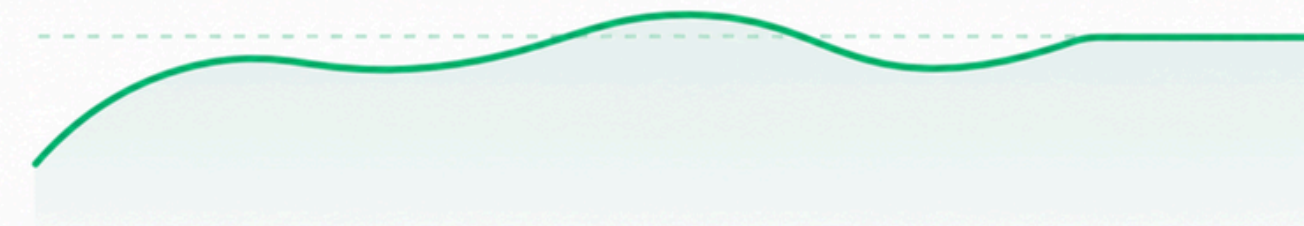
Virtual Users

50

Total Requests

Throughput Curve (Target 8.3 req/s)

10s Window



Test Environment : API performance and system responses were validated using Google Cloud Console under controlled test conditions.

Load Test : 50 API requests were executed across 5 concurrent users to evaluate system stability and response handling.



Project Deliverables & Links



GitHub Repository

- Contains the full production-ready source code and project repository.
- Includes complete system architecture, database schema, and API specs.
- Features a detailed README.md with local setup and build instructions.



[GitHub Link](#)



Demo Video (3 Minutes)

- A concise, step-by-step video walkthrough of our core application workflow.
- Demonstrates real-world usage scenarios tailored for Karnataka State Police.
- Explains the key features, technology stack, and business impact visually.



[Demo Video](#)



Deployed Link

- Live Demo: Fully integrated app with real-time APIs and live database connectivity.
- **Login** : kabilanka.cs24@bitsathy.ac.in
Pass : Queenking@257



[Dashboard URL](#)

Demo PII Access Instructions

Officer Name: Any name

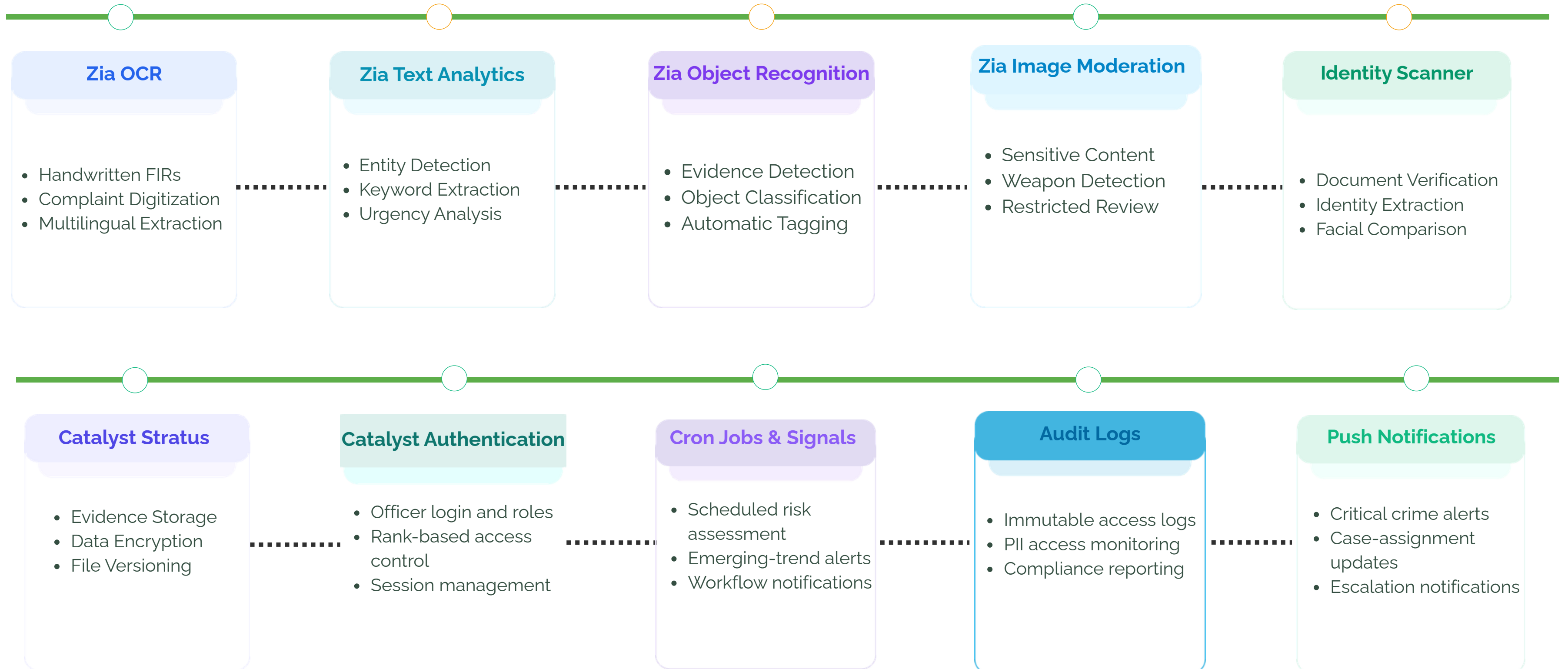
KGID / Badge Number: KG#####

Assigned Station: Command Center

Command Center provides access to all demonstration cases. Selecting a police station displays only cases assigned to that station.



Future Development





Karnataka State Police
Government of Karnataka

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THANK YOU